

Luis Gerardo García Martínez

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EDUCATION

Monterrey Institute of Technology

B.S. Chemical Engineering

- GPA: 97/100

Monterrey, México

Aug 2024 - July 2028

High School: PrepaTec Eugenio Garza Lagüera

Honorable Mention / IB French Course H.L.

Monterrey, México

Aug. 2021 - May 2024

EXPERIENCE

Undergraduate Research Student

Tecnológico de Monterrey

October 2024 – Present

Monterrey, México

- Developed numerical simulations in MATLAB to model structured-light propagation and nonlinear interactions with chiral molecules.
- Explored theoretical frameworks relating structured-light polarization to enantiomer discrimination.
- Researched and designed optical configurations to optimize the experimental response for chiral discrimination.

Undergraduate Research Intern

Imperial College London – Surfaces and Particle Engineering Laboratory

June 2026 – Aug. 2026

London, United Kingdom

- Researched the effect of L-arginine concentration as a soft template on vancomycin crystallisation using vapour diffusion screening experiments.
- Evaluated different buffer concentrations to determine their influence on vancomycin crystal formation, morphology, and experimental reproducibility.
- Programmed and operated Opentrons OT-2 liquid-handling robot to automate solution preparation and systematic crystallisation experiments.

Undergraduate Research Intern

Colgate University

June – Aug. 2025

Hamilton, New York

- Assembled an experimental setup for Sum Frequency Generation Circular Dichroism (SFG-CD) using a bichromatic laser beam.
- Collaborated with faculty members to identify and optimize the optical configuration used for SFG-CD measurements.
- Crystallized mandelic acid enantiomers through hot filtration to prepare samples for SFG-CD analysis.

EXTRACURRICULAR ACTIVITIES

AIChE Chem-E-Cube Competition Team

Business Team Member, Tecnológico de Monterrey

2026 – Present

Monterrey, México

- Contributed to the development of the business plan for a one-cubic-foot carbon dioxide capture system designed for the AIChE Chem-E-Cube Competition.
- Conducted market and implementation research to identify potential customers, strategic locations, and commercialization opportunities for the carbon capture technology.

AIChE Regional Student Paper Competition

Southwest AIChE Regional Conference, Texas Tech University

March 2026

Lubbock, Texas

- Awarded **first place** in the Regional Student Paper Competition for delivering a 20-minute technical presentation at the 2026 Southwest AIChE Regional Conference.
- Presented the research project “First Steps toward the Characterization of Enantiomers with Sum Frequency Generation Circular Dichroism (SFG-CD).”

AICHE Undergraduate Student Poster Competition

Hynes Convention Center

November 2025

Boston, Massachusetts

- Presented the research poster “First Steps toward the Characterization of Enantiomers with Sum Frequency Generation Circular Dichroism (SFG-CD)” at the 2025 AIChE Annual Meeting in the Materials Engineering and Sciences Division.

Chemical Engineering Student Society

Tecnológico de Monterrey

July 2025 – Present

Monterrey, México

- Served as **Academic Affairs Coordinator** during the 2025–2026 academic year, supporting the planning and execution of academic initiatives for Chemical Engineering students.
- Appointed **Academic Affairs Director** in July 2026, leading the planning and coordination of academic activities and student development initiatives.

Chemical Engineering Professional Development Program - Vice President

Tecnológico de Monterrey

July 2026 – Present

Monterrey, México

- Organize company visits that expose students to industrial applications and career opportunities in Chemical Engineering.
- Coordinate workshops and technical courses related to Chemical Engineering and students professional development.

Human Resources Director

AIIESEC in Monterrey

Aug. 2025 – Nov. 2025

Monterrey, Nuevo León

- Led the recruitment and selection process for open positions across the organization’s operations in Nuevo León, México.
- Coordinated organizational events and activities to support member engagement and team development.
- Reviewed the status, participation, and activity of members across the regional organization.

PROJECTS

Circularly Polarized Laguerre–Gaussian Beams in Turbulent Media

January 2026 – Present

- Developed MATLAB simulations to model the propagation and transmission of circularly polarized Laguerre–Gaussian beams through turbulent optical media.
- Evaluated whether circular polarization improves beam transmissivity by comparing transmitted optical power under different turbulence conditions and polarization states.

First Steps toward the Characterization of Enantiomers with SFG-CD

October 2024 – Present

- Studied enantiomer discrimination using nonlinear optics by combining Second Harmonic Generation and SFG for the characterization of mandelic acid.
- Designed and implemented an SFG-CD experiment to investigate second-order nonlinear optical interactions for chiral molecule analysis.
- Analyzed preliminary numerical and experimental results describing the SFG-CD response of mandelic acid enantiomers.

TECHNICAL SKILLS

Computational Tools: Python, MATLAB, Aspen/HYSYS, COMSOL Multiphysics, LaTeX.

Laboratory Automation: Opentrons OT-2 liquid-handling robot.

Optical Instrumentation: Nd:YAG GCE-100 laser at 1064 nm and 532 nm, helium-neon lasers, spatial light modulators, dichroic mirrors, and polarization optics.

Laboratory Techniques: Vapour diffusion crystallisation, hot-filtration crystal growth, solution preparation, beam alignment, polarization control, XRD, and XAS.